Application 1: Horizontal Elastic Pendulum Solution

1. Nature of Motion

The graph shows the position x oscillating sinusoidally with time. The amplitude and period are constant. Thus, the motion is **Simple Harmonic Motion (SHM)**.

2. Type and Regime of Oscillations

- **Type (Mode):** Free, Undamped Oscillations.
  - *Free:* No external driving force is acting on the system after $t = 0$.
  - *Undamped:* The amplitude X_m remains constant ($6\ \text{cm}$) over time, implying no energy dissipation (friction is negligible).
- **Regime:** Periodic (Sinusoidal).

3. Expression of Mechanical Energy (ME)

For a horizontal elastic pendulum, the gravitational potential energy is constant (can be taken as zero at the horizontal reference). The Mechanical Energy is the sum of Kinetic Energy (KE) and Elastic Potential Energy (PE_{el}).

$$ME = KE + PE_{el}$$

$$ME = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

Where:

- m is the mass of the block.
- v is the velocity ($v = \dot{x}$).
- k is the stiffness constant of the spring.
- x is the elongation/compression from equilibrium.

4. Differential Equation

Since friction is negligible, the Mechanical Energy is conserved:

$$\frac{d(ME)}{dt} = 0$$

Differentiating the energy expression with respect to time:

$$\frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2\right) = 0$$

$$\frac{1}{2}m(2v\dot{v}) + \frac{1}{2}k(2x\dot{x}) = 0$$

Since $v = \dot{x}$ and $\dot{v} = \ddot{x}$:

$$mv\ddot{x} + kxv = 0$$

Factor out v (where $v \neq 0$ generally):

$$v(m\ddot{x} + kx) = 0$$

This yields Newton's Second Law form:

$$m\ddot{x} + kx = 0$$

Dividing by m, we get the standard differential equation:

$$\ddot{x} + \frac{k}{m}x = 0$$

5. Expression of Natural Pulsation (ω_0)

The general differential equation for SHM is:

$$\ddot{x} + \omega_0^2 x = 0$$

Comparing this with our derived equation $\ddot{x} + \frac{k}{m}x = 0$, we identify:

$$\omega_0^2 = \frac{k}{m}$$

Therefore:

$$\omega_0 = \sqrt{\frac{k}{m}}$$

6. Determination of Constants from Graph

We are given the general form $x(t) = X_m \sin(\omega_0 t + \varphi)$.

1. **Amplitude (X_m):** From the graph, the maximum displacement is 6 cm.

$$X_m = 6 \text{ cm} = 0.06 \text{ m}$$

2. **Period (T) and Pulsation (ω_0):** The oscillation repeats every 1.0 s (e.g., crossing $x = 0$ downwards at $t = 0$ and again at $t = 1$).

$$T = 1.0 \text{ s}$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{1} = 2\pi \text{ rad/s}$$

3. **Phase Constant (φ):** At $t = 0$, the graph shows $x(0) = 0$. Substituting into the equation:

$$x(0) = X_m \sin(\omega_0(0) + \varphi) = 0$$

$$\sin(\varphi) = 0 \Rightarrow \varphi = 0 \text{ or } \pi$$

Checking velocity: The slope of the graph at $t = 0$ is negative (the curve goes down).

$$v(t) = \dot{x}(t) = X_m \omega_0 \cos(\omega_0 t + \varphi)$$

$$v(0) = X_m \omega_0 \cos(\varphi)$$

For $v(0) < 0$, we need $\cos(\varphi)$ to be negative.

- If $\varphi = 0, \cos(0) = 1 > 0$ (Incorrect).
- If $\varphi = \pi, \cos(\pi) = -1 < 0$ (Correct).

Thus, $\varphi = \pi$ rad.

Time Equation of Motion:

$$x(t) = 0.06 \sin(2\pi t + \pi) \quad \text{(SI units)}$$

(Note: $\sin(\theta + \pi) = -\sin(\theta)$, so this can also be written as $x(t) = -0.06 \sin(2\pi t)$)

7. Calculate Stiffness (k)

Given $m = 160 \text{ g} = 0.16 \text{ kg}$. Using $\omega_0 = \sqrt{\frac{k}{m}}$:

$$k = m\omega_0^2$$

$$k = 0.16 \times (2\pi)^2$$

$$k = 0.16 \times 4\pi^2 \approx 0.16 \times 39.48$$

$$k \approx 6.32 \text{ N/m}$$

(Approximating $\pi^2 \approx 10$ yields $k = 6.4 \text{ N/m}$)

8. Velocity-Time Equation and V_{max}

$$v(t) = \frac{dx}{dt} = \frac{d}{dt}[0.06 \sin(2\pi t + \pi)]$$

$$v(t) = 0.06(2\pi) \cos(2\pi t + \pi)$$

$$v(t) = 0.12\pi \cos(2\pi t + \pi) \text{ m/s}$$

Maximum Velocity (V_{max}): The amplitude of the velocity function is 0.12π.

$$V_{max} = 0.12\pi \approx 0.38 \text{ m/s}$$

9. First Passage through Equilibrium (O)

The particle starts at O ($x = 0$) at $t = 0$. It passes O again when $x(t) = 0$:

$$\sin(2\pi t + \pi) = 0$$

$$2\pi t + \pi = k\pi \quad (k \text{ is an integer})$$

The first time after $t = 0$ corresponds to the half-period. From the graph, the curve crosses the t-axis again at:

$$t = 0.5 \text{ s}$$

10. First Moment Speed Becomes Nil (t_1)

Speed is zero when the object is at a maximum extremum ($x = \pm X_m$). From the graph, the first peak/trough occurs at:

$$t_1 = 0.25 \text{ s}$$

(At this point, $x = -6 \text{ cm}$).

11. Speed at $x = 3 \text{ cm}$ in Negative Direction

We need to find v when $x = 0.03 \text{ m}$ and the direction is negative ($v < 0$).

Method 1: Conservation of Energy

$$ME = \text{constant} \Rightarrow ME_{max} = ME_{instant}$$

$$\frac{1}{2}kX_m^2 = \frac{1}{2}kx^2 + \frac{1}{2}mv^2$$

Multiply by 2 and divide by m (recalling $k/m = \omega_0^2$):

$$\omega_0^2 X_m^2 = \omega_0^2 x^2 + v^2$$

$$v^2 = \omega_0^2 (X_m^2 - x^2)$$

$$v = \pm\omega_0 \sqrt{X_m^2 - x^2}$$

Since the direction is negative, we take the negative root:

$$v = -2\pi\sqrt{0.06^2 - 0.03^2}$$

$$v = -2\pi\sqrt{0.0036 - 0.0009}$$

$$v = -2\pi\sqrt{0.0027} \approx -2\pi(0.052)$$

$$v \approx -0.326 \text{ m/s}$$

$$(\text{Exact: } -3\pi\sqrt{3} \times 10^{-2} \text{ m/s})$$

Method 2: Time Equations

Find t when $x = 3 \text{ cm}$ and $v < 0$.

$$x(t) = -6\sin(2\pi t) = 3$$

$$\sin(2\pi t) = -0.5$$

The reference angle is $\pi/6$. Sine is negative in Quadrants III and IV.

- **Q3:** $2\pi t = \pi + \pi/6 = 7\pi/6$. Checking velocity: $v \propto -\cos(7\pi/6)$. $\cos(7\pi/6)$ is negative, so $-\cos$ is positive. ($v > 0$, Incorrect direction).
- **Q4:** $2\pi t = 2\pi - \pi/6 = 11\pi/6$. Checking velocity: $v \propto -\cos(11\pi/6)$. $\cos(11\pi/6)$ is positive, so $-\cos$ is negative. ($v < 0$, Correct direction).

Using the phase $2\pi t + \pi = 11\pi/6 + \pi = 17\pi/6$ in the velocity equation:

$$v = 0.12\pi\cos(17\pi/6)$$

$$v = 0.12\pi(-\frac{\sqrt{3}}{2})$$

$$v = -0.06\pi\sqrt{3} \approx -0.326 \text{ m/s}$$

12. Shape of Velocity vs Time Curve

See the attached Python generated graph. The velocity curve is a cosine function shifted by π (or simply a negative cosine relative to the phase of sine).

- It starts at a negative maximum: $v(0) \approx -0.38 \text{ m/s}$.
- It crosses zero at extrema of position ($t = 0.25, 0.75$).
- It has the same period $T = 1.0 \text{ s}$.